

AthDGC · Athens-PROIEL

An open diachronic Greek treebank with Indo-European parallels

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The **Athens Diachronic Glossa Chronos** (*Athens-PROIEL*) is an open PROIEL-style dependency treebank spanning Greek diachrony with Indo-European parallels.

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AthDGC is **both** a continuously-updated **PROIEL XML 2.0** treebank **and** an open-source **computational diachronic-linguistics toolkit**:

- LightSIDE-compatible feature extraction + classifier training
- NoSketch-style concordancer
- Fine-tuned Stanza checkpoints (Byzantine, Late-Byzantine, Modern adaptations)
- Argument-structure extractor
- Cross-lingual alignment viewer (Neo4j)
- Quarto template pack for any DH project (this very deck is built with it)

Covers:

- Homeric and Archaic Greek (Homer, Hesiod, Sappho, Pindar)
- Classical (Aeschylus, Sophocles, Euripides, Aristophanes, Plato, Aristotle)
- Koine and NT (Plutarch, Strabo, Lucian, the New Testament)
- Late Antique (Eusebius, Basil, Chrysostom, Procopius)
- Byzantine (Psellos, Anna Komnene, John of Damascus, Niketas Choniates)

with **NT verse-level cross-alignment** to Latin (Vulgate), Gothic (Wulfila), and Old Church Slavonic (Marianus).

Metric	Value
Total corpus rows	89.9 M
Annotated Greek rows	4.08 M
NT-aligned Greek verses	6,861
IE cross-aligned witnesses	4

1. **Discovery** - daily harvest from archive.org, Perseus, First1K, Wikisource, Diorisis
2. **Filtering** - Greek-script ratio, apparatus-criticus rejection, dedup
3. **Conversion** - PROIEL XML 2.0
4. **Annotation** - Stanza `grc_proiel` + analogous models per parallel language
5. **Argument-structure** - subject, object, oblique, voice, aspect per verb
6. **Cross-alignment** - LaBSE + AwesomeAlign (mBERT)
7. **Storage** - PROIEL XML + JSONL + Qdrant + Neo4j

- **Retranslation** of influential texts (Iliad, NT, Septuagint Psalms, classical historiography)
- **Retelling** chains across periods and languages
- **Argument structure** under retranslation

- Lavidas (PI) · Nikiforidou · Haug (Oslo) · Kulikov (Ghent)
- Geka · Symeonidis · Michalareas
- Chionidi · Tsiropina · Plakoutsi · Argyropoulos

NKUA · Athens Digital Glossa Chronos Research Network

Lavidas, N., Nikiforidou, K., Haug, D., Kulikov, L., Geka, V., Symeonidis, V., Michalareas, T., Chionidi, S., Tsiropina, A., Plakoutsi, E., Argyropoulos, E., and the Athens Digital Glossa Chronos Research Network (2026). *AthDGC: Athens Diachronic Glossa Chronos*. Zenodo.

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- Showcase: <https://athdgc.github.io>
- Press: https://athdgc.github.io/athdgc_press.html
- GitHub: <https://github.com/AthDGC>
- HFRI project site: <https://athdgc.github.io/hfri2024>

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